

UNIT-3 ALGEBRA

CASE BASED QUESTIONS

1.

A scholarship is a sum of money provided to a student to help him or her pay for education. Some students are granted scholarships based on their academic achievements, while others are rewarded based on their financial needs.



Every year a school offers scholarships to girl children and meritorious achievers based on certain criteria. In the session 2022 – 23, the school offered monthly scholarship of ₹ 3,000 each to some girl students and ₹ 4,000 each to meritorious achievers in academics as well as sports.

In all, 50 students were given the scholarships and monthly expenditure incurred by the school on scholarships was ₹ 1,80,000.

Based on the above information, answer the following questions :

- (i) Express the given information algebraically using matrices.
- (ii) Check whether the system of matrix equations so obtained is consistent or not.
- (iii) (a) Find the number of scholarships of each kind given by the school, using matrices.
- (b) Had the amount of scholarship given to each girl child and meritorious student been interchanged, what would be the monthly expenditure incurred by the school ?

2. Gautam buys 5 pens, 3 bags and 1 instrument box and pays a sum of ₹ 160. From the same shop, Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays a sum of ₹ 190. Also Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays a sum of ₹ 250.

Based on the above information, answer the following questions :

(I) Convert the given above situation into a matrix equation of the form $AX = B$.

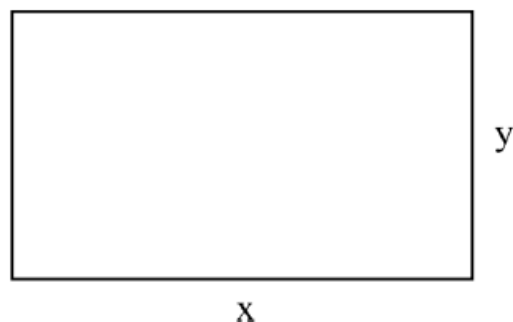
(II) Find $|A|$.

(III) Find A^{-1} .

OR

(III) Determine $P = A^2 - 5A$.

3. An architect is developing a plot of land for a commercial complex. When asked about the dimensions of the plot, he said that if the length is decreased by 25 m and the breadth is increased by 25 m, then its area increases by 625 m^2 . If the length is decreased by 20 m and the breadth is increased by 10 m, then its area decreases by 200 m^2 .



On the basis of the above information, answer the following questions :

- (i) Formulate the linear equations in x and y to represent the given information.
- (ii) Find the dimensions of the plot of land by matrix method.

4. Three students, Neha, Rani and Sam go to a market to purchase stationery items. Neha buys 4 pens, 3 notepads and 2 erasers and pays ₹ 60. Rani buys 2 pens, 4 notepads and 6 erasers for ₹ 90. Sam pays ₹ 70 for 6 pens, 2 notepads and 3 erasers.

Based upon the above information, answer the following questions :

- (i) Form the equations required to solve the problem of finding the price of each item, and express it in the matrix form $AX = B$. 1
- (ii) Find $|A|$ and confirm if it is possible to find A^{-1} . 1
- (iii) (a) Find A^{-1} , if possible, and write the formula to find X . 2
- OR**
- (iii) (b) Find $A^2 - 8I$, where I is an identity matrix. 2